

NANYANG TECHNOLOGICAL UNIVERSITY
SEMESTER 2 EXAMINATION 2022-2023
EE6403 – DISTRIBUTED MULTIMEDIA SYSTEMS

April / May 2023

Time Allowed: 3 hours

INSTRUCTIONS

1. This paper contains 5 questions and comprises 4 pages.
2. Answer all 5 questions.
3. All questions carry equal marks.
4. This is a closed book examination.
5. Unless specifically stated, all symbols have their usual meanings.

1. In an image compression scheme, Singular Value Decomposition (SVD) is used to compress an image block $\mathbf{P}$ given by:

$$\mathbf{P} = \begin{bmatrix} 1 & 2 & 1 \\ 4 & 3 & 2 \end{bmatrix}.$$

The SVD of $\mathbf{P}$ is given as:

$$\mathbf{P} = \mathbf{U}\mathbf{\Sigma}\mathbf{V}^T.$$

- (a) Write down the characteristic equation used to compute the singular values in $\mathbf{\Sigma}$. Hence, find the matrix $\mathbf{\Sigma}$.
(6 Marks)
- (b) Find the matrix $\mathbf{U}$.
(6 Marks)
- (c) Given that $\mathbf{V} = [\mathbf{v}_1 \quad \mathbf{v}_2 \quad \mathbf{v}_3]$ where $\mathbf{v}_1$, $\mathbf{v}_2$, and $\mathbf{v}_3$ are column vectors, find the vector $\mathbf{v}_1$.
(4 Marks)

Note: Question No. 1 continues on page 2.

- (d) SVD is used to compress two image blocks **Q** and **R** below separately using low-rank representation:

$$\mathbf{Q} = \begin{bmatrix} 1 & 1 & 1 & 2 & 2 \\ 4 & 2 & 1 & 1 & 1 \\ 2 & 2 & 2 & 4 & 4 \\ 8 & 4 & 2 & 2 & 2 \end{bmatrix}, \mathbf{R} = \begin{bmatrix} 2 & 2 & 1 & 2 & 2 \\ 1 & 1 & 2 & 1 & 1 \\ 1 & 2 & 3 & 2 & 1 \\ 3 & 3 & 2 & 2 & 0 \end{bmatrix}.$$

A user decides to use a rank of 2 to compress these two image blocks. State clearly which of the two decompressed image blocks will have a larger reconstruction error. Justify your answer briefly.

(4 Marks)

2. (a) Draw a diagram of the baseline JPEG decoder and clearly label all the key components. State which key step in the baseline JPEG compression results in the information loss during image compression.

(8 Marks)

- (b) A user would like to apply the JPEG compression to medical images. Suggest a suitable JPEG mode to compress the medical images, and briefly justify your answer.

(4 Marks)

- (c) A user is investigating different possible ways to improve the performance of the baseline JPEG compression. State clearly whether you agree or disagree with the following suggestions and briefly discuss the practicality/usefulness of the suggestions.

(i) Suggestion A: remove the quantization step in the baseline JPEG compression.

(ii) Suggestion B: replace the Discrete Cosine Transform (DCT) with the Karhunen-Loeve Transform (KLT) in the transform step of the baseline JPEG compression.

(8 Marks)

3. (a) In an application, a user would like to compress a video using two different methods. The video has the following format:

Resolution of the luminance plane: 720×480 pixels

Color depth of each pixel in each luminance and chrominance plane: 8 bits/pixel

Chroma subsampling format is 4:2:0

Frame rate: 30 frames per second

Note: Question No. 3 continues on page 3.

- (i) In method 1, the user would like to compress the video using the MPEG-2 standard. The settings are given as follows.

The Group-of-Pictures (GOP) structure used is IBBPBBPBB. The average compression ratios for the I-frames, P-frames, and B-frames are assumed to be 10:1, 20:1, and 40:1, respectively.

Calculate the average bitrate of the compressed video in bits/s. State any assumption(s) you make in the calculation.

- (ii) In method 2, the user would like to compress the video using the Motion JPEG standard. The Motion JPEG standard compresses each video frame independently as a JPEG image. The average compression ratio for a JPEG image in the video is assumed to be 10:1.

Calculate the average bitrate of the compressed video in bits/s. State any assumption(s) you make in the calculation.

- (iii) Write down the advantage(s) and disadvantage(s) of using the MPEG-2 standard over the Motion JPEG standard in the video compression.

(14 Marks)

- (b) With the aid of a diagram, briefly explain the display operation of a Liquid Crystal Display (LCD) TV.

(6 Marks)

4. (a) A Cyclic Redundancy Check (CRC) system uses a generator polynomial of $P(X) = X^4 + X + 1$. A data sequence 10110011 (most significant bit on the left) is to be coded and transmitted using the CRC.

- (i) Briefly explain the objective of CRC in data transmission.

- (ii) Determine the transmitted codeword.

- (iii) Suppose transmission errors occur in the second and third bits (from the left) of the transmitted codeword in part (a)(ii). Can the CRC detect the transmission errors? Justify your answer briefly.

(10 Marks)

- (b) Briefly discuss the main difference(s) between Frequency Division Multiple Access (FDMA) and Code Division Multiple Access (CDMA).

(5 Marks)

- (c) State clearly which of the following: (i) Transmission Control Protocol (TCP) or (ii) User Datagram Protocol (UDP) can provide better reliability in data transmission over the networks. Justify your answer briefly.

(5 Marks)

5. (a) A particular Convolutional Neural Network (CNN) consists of several convolutional+RELU layers, pooling layers, Fully Connected (FC) layers, and followed by a Softmax layer. The last FC layer of the CNN has the following values for a training sample:

Input to the last FC layer: $\mathbf{x} = [2 \ 2 \ 3 \ 1 \ 0]^T$,

Last FC layer weight matrix: $\mathbf{W} = \begin{bmatrix} 0.1 & 0.1 & 0.1 & 0.2 & 0.2 \\ 0.4 & 0.3 & 0.2 & 0.1 & 0.1 \\ 0.1 & 0.1 & 0.1 & 0.1 & 0.1 \end{bmatrix}$,

Last FC layer bias: $\mathbf{b} = [1 \ -1 \ 1]^T$.

- (i) Find the output $\mathbf{y}$ of the last FC layer.
- (ii) Find the output $\mathbf{p}$ after the Softmax layer.
- (iii) Assume that the ground truth output label of this training sample is given by $\mathbf{t} = [1 \ 0 \ 0]^T$, calculate the Softmax loss for this training sample.

(10 Marks)

- (b) A user would like to use transfer learning to classify X-ray images with 5 output categories. The user chooses a VGG model that has been pretrained using ImageNet dataset with 1000 output categories. The user then performs transfer learning based on the pretrained network to classify the X-ray images. The distribution of the X-ray training images is given in Table 1 below.

Table 1

Category	1	2	3	4	5
Number of X-ray training images	200	400	300	10	20

- (i) List the key steps in the transfer learning process.
- (ii) Briefly discuss the potential problem(s) in using transfer learning for this classification problem.

(10 Marks)

END OF PAPER

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Please read the following instructions carefully:

- 1. Please do not turn over the question paper until you are told to do so. Disciplinary action may be taken against you if you do so.**
2. You are not allowed to leave the examination hall unless accompanied by an invigilator. You may raise your hand if you need to communicate with the invigilator.
3. Please write your Matriculation Number on the front of the answer book.
4. Please indicate clearly in the answer book (at the appropriate place) if you are continuing the answer to a question elsewhere in the book.